VEDANG TRIVEDI

Software Engineer | Data Engineer | Distributed Systems, Python & ML Systems

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Profile Summary

Software Engineer with experience managing application development lifecycles from requirement gathering to final delivery. Designed, built, documented, and shipped two Power Platform applications at Kraft Heinz used daily by **50+ people** across **10 towers**, coordinating process automation estimated to save **1,600+ hours** a year. Also builds systems-level research: an entropy-aware LLM-routing model that cuts inference cost by **78%**, and a learned cardinality estimator for query-plan optimization.

Technical Skills

App Dev & Automation: Power Apps, Power Automate, Git/GitHub, Figma, LangChain
Languages: C, C++, Python, SQL, JavaScript
Databases: PostgreSQL, MongoDB, MySQL, Redis
Systems & ML: Distributed Systems, Query Optimization, QLoRA/PEFT Fine-Tuning, XGBoost
Data & BI: Power BI, Pandas/NumPy, NLTK/spaCy

Professional Experience

Kraft Heinz — **Feb 2026 – Present**
KHMS/AIT Intern, GCC India — *Ahmedabad, Gujarat*

- Managed the AIT Power App development lifecycle end-to-end (5 screens, 20 SharePoint columns across 4 data types) from requirement gathering and process mapping to final delivery, deploying a standardized 7-stage lifecycle model to **50+ users** across **10 towers**.
- Implemented 6+ conditional logic rules (field visibility, record locking, cascading filters, form validation) and 2 automated email-notification triggers, streamlining progress tracking and reporting to save an estimated **1,130+ hours a year** of manual effort.
- Monitored project quality and resolved **12 documented defects** through root-cause analysis across iterative build-test-fix cycles with users, delivering the project on time using Microsoft 365 licensing (Power Apps, SharePoint, Power Automate) with zero external budget.
- Coordinated the development and documentation of the Health Check and OpEx Survey app suite on a governed SharePoint backend (**5 lists**) – wizard-based flow, dependent dropdowns, submission validation – for **50+ users** across **11 towers**, ensuring quality delivery that cut submission errors to near-zero and saved an estimated **267 hours a year**, with role-based access control for governance.

Karma Technolabs — **Jan 2024 – Apr 2024**
Web Development Intern — *Ahmedabad, Gujarat*

- Enhanced search, filter, and upload functionality on a live web CRM platform, coordinating across the complete development-to-deployment lifecycle to deliver projects within defined timelines.

Projects

UGAD-Lite: Uncertainty-Guided Adaptive Distillation | *ML Systems*

- Engineered an entropy-conformal router that dynamically directs queries between a large LLM (GPT-4o) and a fine-tuned Small Language Model (Phi-3-Mini via QLoRA), calibrated with conformal prediction for statistical safety guarantees.
- Cut LLM inference cost and token usage by 78% at a 94.8% task success rate on GSM8K and synthetic arithmetic benchmarks, with a <5% false-negative rate.

CardEst | *SQL, Query Optimization*

- Built an XGBoost-based cardinality estimator trained on the TPC-H dataset to improve query-optimizer execution-plan accuracy.
- Cut query latency and resource usage in 70% of evaluated cases by improving Pearson correlation and reducing median estimation error.

Raft Census (DBFS) | *Distributed Systems, Raft Consensus*

- Built a language-agnostic, decentralized file-storage system using Raft cluster consensus (Redis + MongoDB-backed), with work-stealing task distribution to support concurrent multi-client create/read/write/delete operations with efficient search and caching.

UML2PlantUML | *LLMs, Machine Learning*

- Built a small-scale, local reproduction of the approach from Sadeghi et al. (MODELS 2026, arXiv:2607.28987) for handwritten UML-to-PlantUML generation, preprocessing a dataset of activity and sequence diagram images for model training.
- Fine-tuned and benchmarked 7 vision-language models (ViT-BART, BLIP, Flan-T5, ViT-GPT2, Pix2Struct, CLIP, MiniCPM) to generate PlantUML code from class diagram images, and deployed the trained model on Kaggle, where it has drawn numerous downloads.

Distributed Database System | *2PC, Query Optimization*

- Engineered a distributed database with a coordinator-worker topology, implementing horizontal/vertical data fragmentation and a global data dictionary for location-transparent query routing.
- Implemented Two-Phase Commit (2PC) for multi-node transaction atomicity and semi-join query optimization, cutting cross-node bandwidth by 60–70% and query latency by 3–4x versus centralized execution.

Education

DA-IICT (Dhirubhai Ambani Institute of Information and Communication Technology) — **2024 – 2026**
M.Tech, ICT (Software Systems) | CPI: 7.47/10 — *Gandhinagar, Gujarat*

Gujarat Technological University — **2020 – 2024**
B.E., Computer Engineering | CGPA: 7.77/10 — *Ahmedabad, Gujarat*

Positions of Responsibility

Teaching Assistant – Software Engineering & Object-Oriented Programming — **Aug 2024 – Present**
DA-IICT, Gandhinagar — *Gandhinagar, Gujarat*

- Communicated complex topics and conducted lab sessions for 100+ students, managing deadlines for designing and evaluating assignments/exams, and mentoring students in Software Engineering and Object-Oriented Programming.

Achievements

- GATE CS 2024 qualifier.